

Title

Adherence Patterns and Outcome Predictors in Digital-Supported Maternal Health Intervention

Abstract

Secondary analysis examined behavioral adherence patterns and outcome prediction in our 378-participant trial. Participants randomized to the intervention showed variable engagement: 68% completed recommended coaching sessions in the interpregnancy period, declining to 54% during the subsequent pregnancy. Device-measured activity consistently showed improvement during intervention periods (mean 25% increase in daily steps during active coaching, $p < 0.001$). Participants achieving weight loss during the interpregnancy period (mean 4.2 kg, $n = 118$) experienced significantly better primary outcomes compared to those without weight loss (26% vs 45% adverse outcomes, $p = 0.003$). Among intervention completers (defined as $\geq 75\%$ engagement), the intervention effect on the composite outcome was more pronounced (adjusted OR 0.44 vs 0.59 for intention-to-treat analysis). Secondary metabolic markers including fasting glucose and triglycerides improved with greater app engagement. Demographic analysis identified maternal age, education, and prior pregnancy complications as predictors of engagement. Health economic analysis suggested cost-savings of approximately 3,200 per participant assigned to the intervention due to reduced pregnancy complications. Results support the feasibility of digital health platforms as adjuncts to routine care, with effect sizes dependent on engagement and adherence.